

★ WEEK 1 · PART 2 of 3

The Stylized Facts of Returns

The Empirical Fingerprints of Financial Markets

Fat tails · volatility clustering · leverage · the predictability split

Across nearly every market and era, returns share a handful of robust statistical patterns. These 'stylized facts' are the targets every model in this course must reproduce.

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PART 2 · STYLIZED FACTS

What Every Model Must Explain

Before we can model returns, we must know what real returns actually look like. This part catalogs the six patterns that show up everywhere.

Why Study Stylized Facts First?

Three reasons they come before any equation

1

They are the benchmark every model is judged against

A model is only useful if it reproduces what we see in the data. The stylized facts define 'what we see'.

2

They are astonishingly universal

The same patterns appear in stocks, bonds, FX, and crypto — across centuries and continents. That universality is itself a clue.

3

They tell us where prediction is possible — and where it is not

The single most important fact splits the world: the direction of returns is unpredictable, but their magnitude is not.

A model that cannot reproduce the stylized facts is wrong, no matter how elegant.

What Is a Stylized Fact?

A robust empirical regularity, not a theorem

A stylized fact is a statistical pattern that reappears across almost all financial return series — a qualitative truth robust enough to build models around.

What qualifies

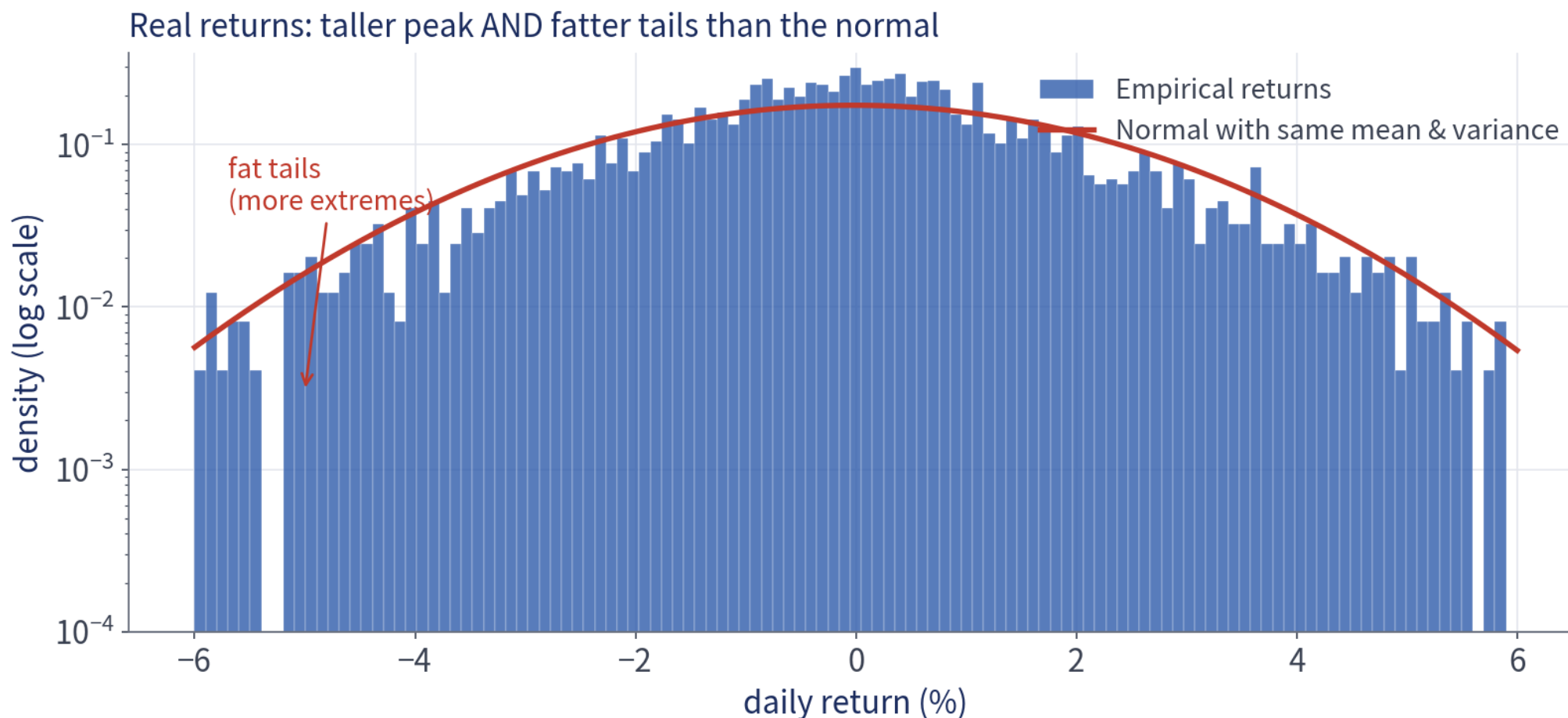
- Seen across many assets, markets, and time periods.
- Survives changes in sample and frequency.
- Qualitative and robust — not a precise numerical law.

How we use them

- As a checklist: does my model reproduce them?
- To rule out models that assume normal, i.i.d. returns.
- To motivate the tools of Weeks 2 onward (GARCH, etc.).

The Distribution of Returns Is Not Normal

Compared with a normal of the same mean and variance, real returns have a taller central peak and much fatter tails.



Describing a Distribution, Part 1: Mean & Variance

The first two moments — center and spread

$$\mu = \mathbb{E}[R], \quad \sigma^2 = \mathbb{E}[(R - \mu)^2], \quad \sigma = \sqrt{\sigma^2}$$

Mean μ — the center

- The average return; where the distribution sits.
- For daily equity returns it is tiny and hard to estimate.

Variance σ^2 / vol σ — the spread

- How widely returns scatter around the mean.
- Volatility σ is the everyday measure of risk.
- Much easier to estimate than the mean — and it moves.

Describing a Distribution, Part 2: Skewness & Kurtosis

The third and fourth moments — asymmetry and tail weight

$$S = \frac{\mathbb{E}[(R-\mu)^3]}{\sigma^3}, \quad K = \frac{\mathbb{E}[(R-\mu)^4]}{\sigma^4}$$

Skewness S — the lean

- $S < 0$: a longer left tail (big losses more likely).
- $S > 0$: a longer right tail.
- Equity returns are typically negatively skewed.

Kurtosis K — the tails

- Measures how much probability sits in the tails.
- Normal has $K = 3$; anything above means fat tails.
- Excess kurtosis = $K - 3$ is the fat-tail gauge.

The Autocorrelation Function (ACF)

Our tool for asking: does the past predict the future?

$$\rho_R(k) = \frac{\text{Cov}(R_t, R_{t-k})}{\text{Var}(R_t)}$$

What it measures

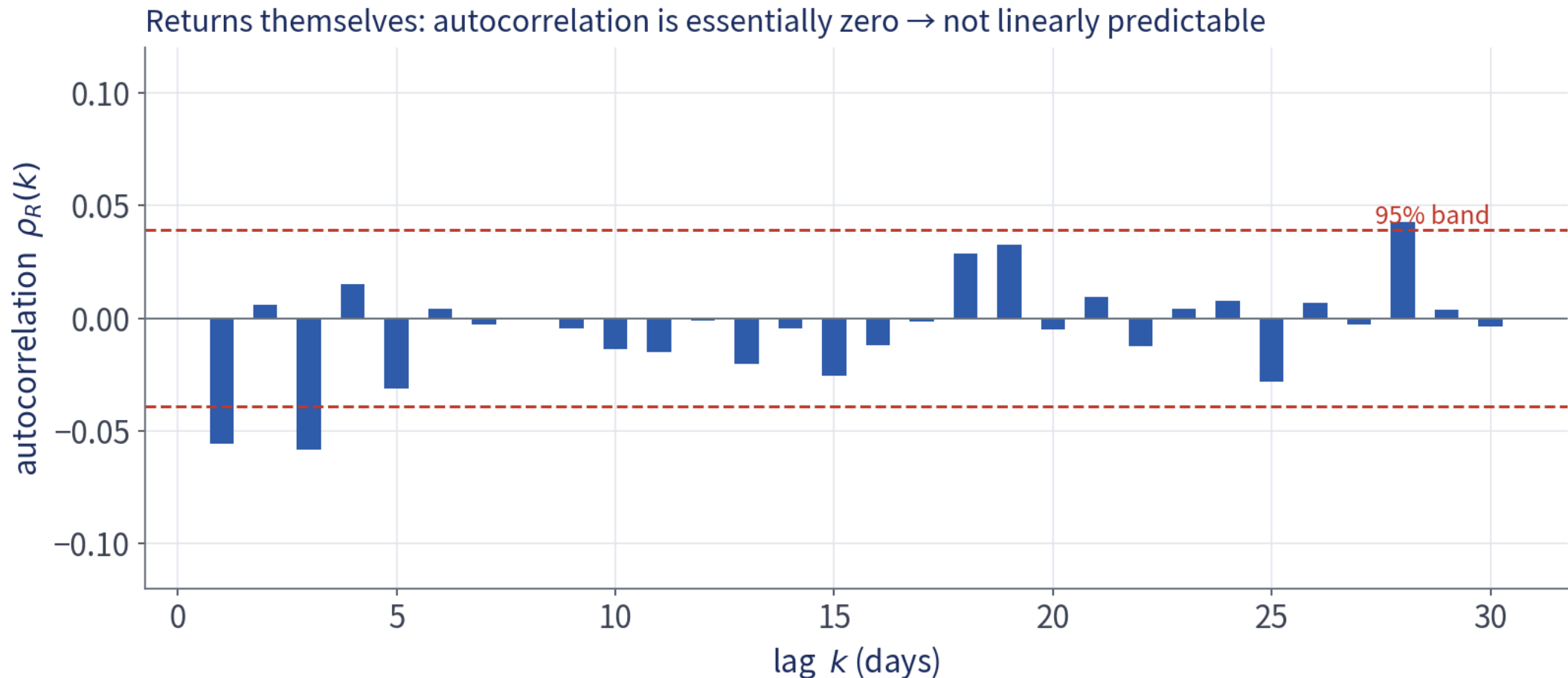
- The correlation of a series with its own past, k steps back.
- $\rho(k)$ near 0 $\rightarrow$ no linear link between today and lag k .
- $\rho(k)$ far from 0 $\rightarrow$ predictable structure at that lag.

How to read the plot

- Bars are $\rho(k)$ at each lag $k = 1, 2, 3, \dots$
- Dashed lines are the 95% 'no-correlation' band.
- Bars inside the band are statistically indistinguishable from 0.

Returns Have Almost No Autocorrelation

Every bar sits inside the 95% band: $\rho_R(k) \approx 0$. The direction of returns is not linearly predictable — the basis of market efficiency (Part 3).



Fat Tails (Excess Kurtosis)

Rare does not mean impossible

$$\text{excess kurtosis} = K - 3 > 0 \implies \text{fat tails}$$

What it means

- Extreme moves happen far more often than a normal predicts.
- A 'once-in-a-century' normal event can arrive every few years.
- This return series has excess kurtosis of about 5.

Why it matters

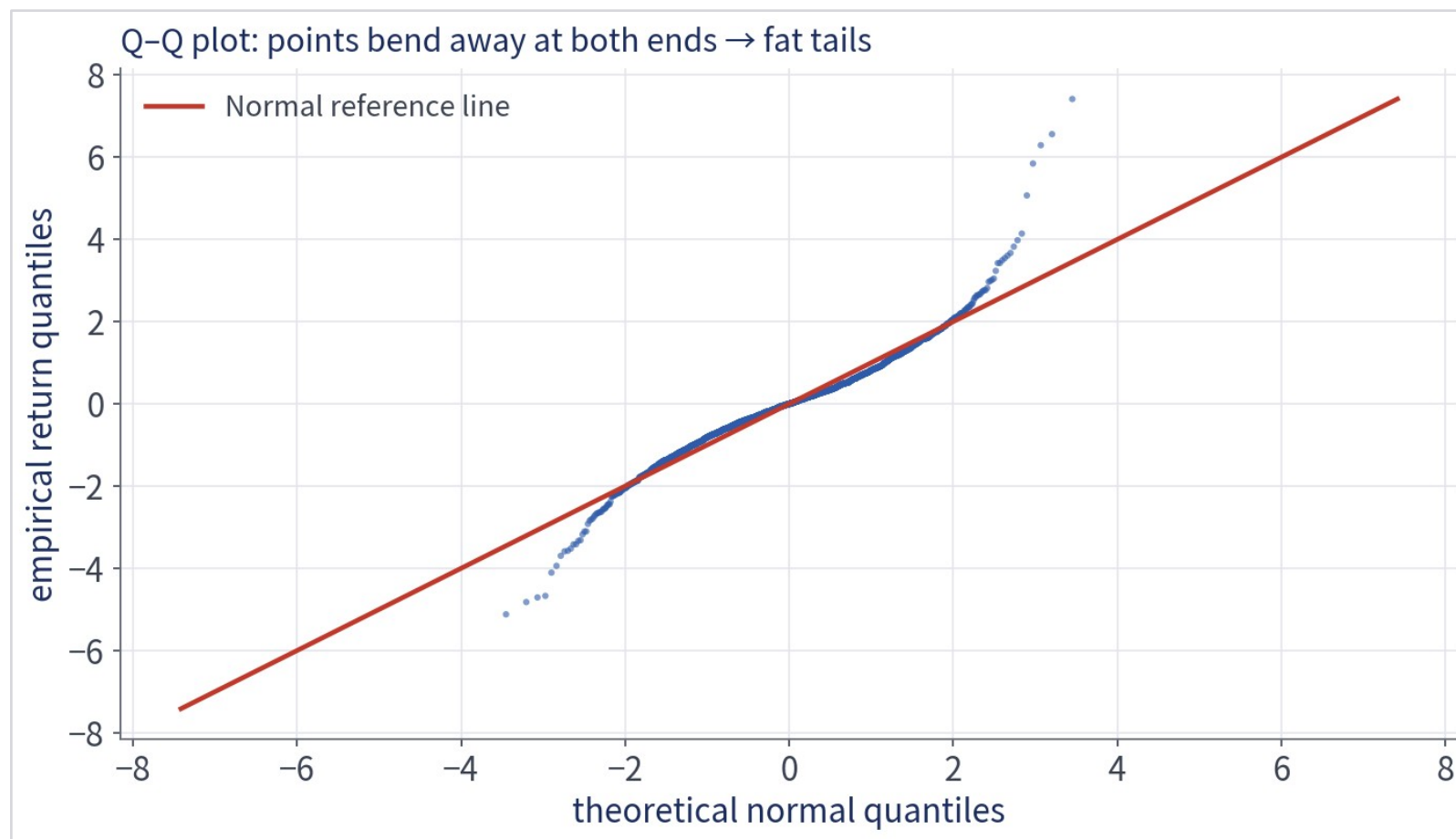
- Risk models built on the normal understate tail risk badly.
- Value-at-Risk and stress tests must use fat-tailed models.
- The next chart makes the fat tails visible.

Seeing Fat Tails: The Q-Q Plot

If returns were normal, the points would lie on the red line.

Reading a Q-Q plot

- Points on line = normal.
- Both ends bend away.
- Low end below the line.
- High end above the line.
- That S-shape = fat tails.



How Heavy Are the Tails?

From a single number to a tail law

Beyond kurtosis, we describe tails by a power law: the probability of an extreme move of size x decays like x raised to a negative tail index α .

The tail index α

- Smaller $\alpha \rightarrow$ heavier tails, more extreme events.
- Equity returns typically show α of roughly 3 to 5.
- Only moments up to order α are finite.

The Student-t link

- A Student-t with ν degrees of freedom has tail index ν .
- Small ν (e.g. 4–6) reproduces realistic fat tails.
- As ν grows, the t converges back to the normal.

Distributions That Capture Fat Tails

Choosing the right model for the tails

Distribution	Tail behavior	Key parameter	Use when
Normal (Gaussian)	Thin — decays very fast	mean, variance	A first approximation, or long horizons
Student-t	Fat — power law	degrees of freedom ν	Modeling realistic daily fat tails
Generalized error / Laplace	Heavier than normal	shape parameter	Peaked center with moderate tails
Empirical / EVT	Data-driven, tail index	threshold, tail index	Estimating extreme quantiles (VaR)

Rule of thumb: start normal for intuition, switch to Student-t (or EVT for the far tail) for real risk work.

Gain / Loss Asymmetry (Negative Skew)

Markets fall faster than they rise

Equity return distributions typically lean left: the left tail is longer than the right. Large losses are more likely than large gains of the same size.

What you observe

- Sharp, deep crashes; slower, steadier climbs.
- The biggest single-day moves are usually down days.
- Skewness S is negative for most equity indices.

Why it happens

- Fear spreads and forced selling feed on itself.
- Deleveraging and margin calls amplify declines.
- Closely tied to the leverage effect (a few slides on).

Volatility Clustering

Turbulence comes in waves

"Large changes tend to be followed by large changes, of either sign, and small changes by small changes." — Mandelbrot, 1963

The signature

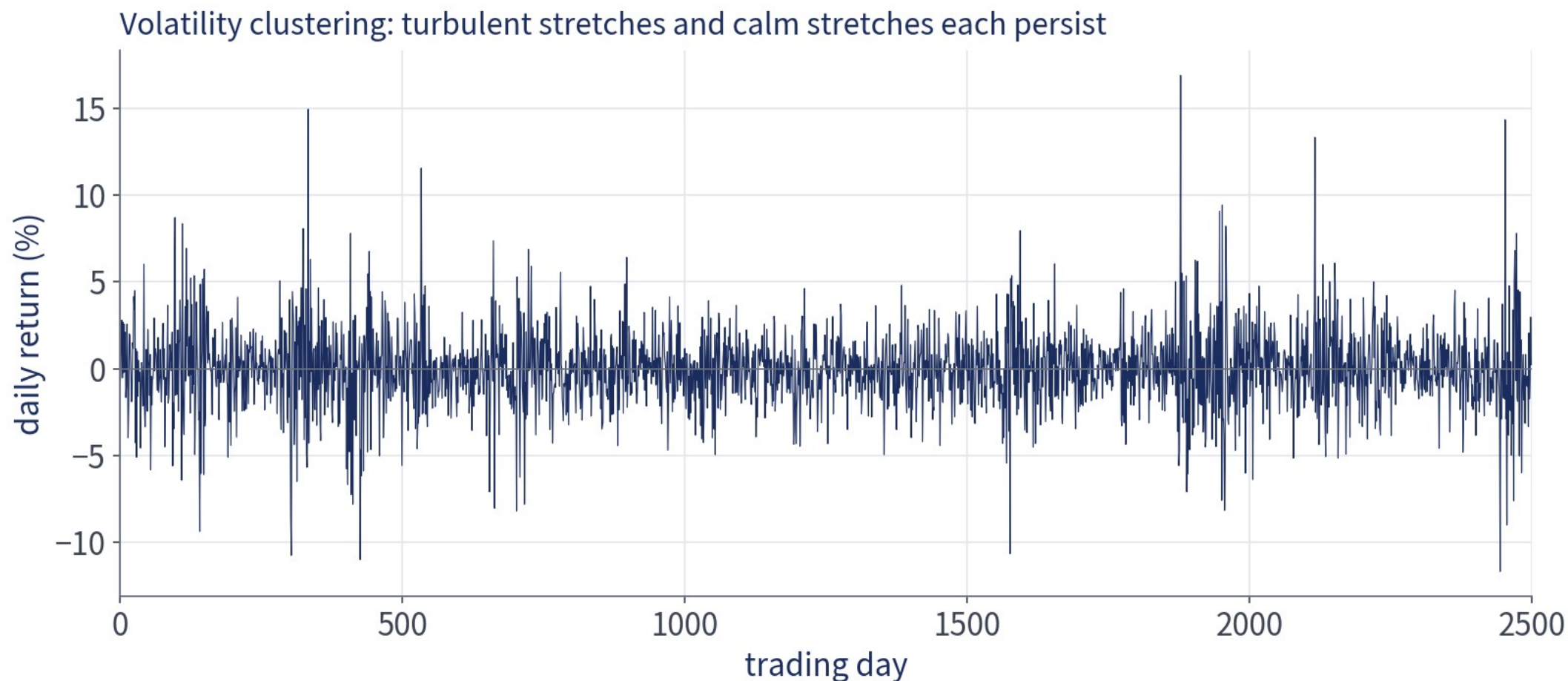
- Calm periods and stormy periods each persist.
- Volatility today is a strong predictor of volatility tomorrow.
- This is what makes risk forecasting possible.

The precise statement

- Returns themselves show no autocorrelation ...
- ... yet their squares (or absolute values) do.
- Two charts make this split unmistakable.

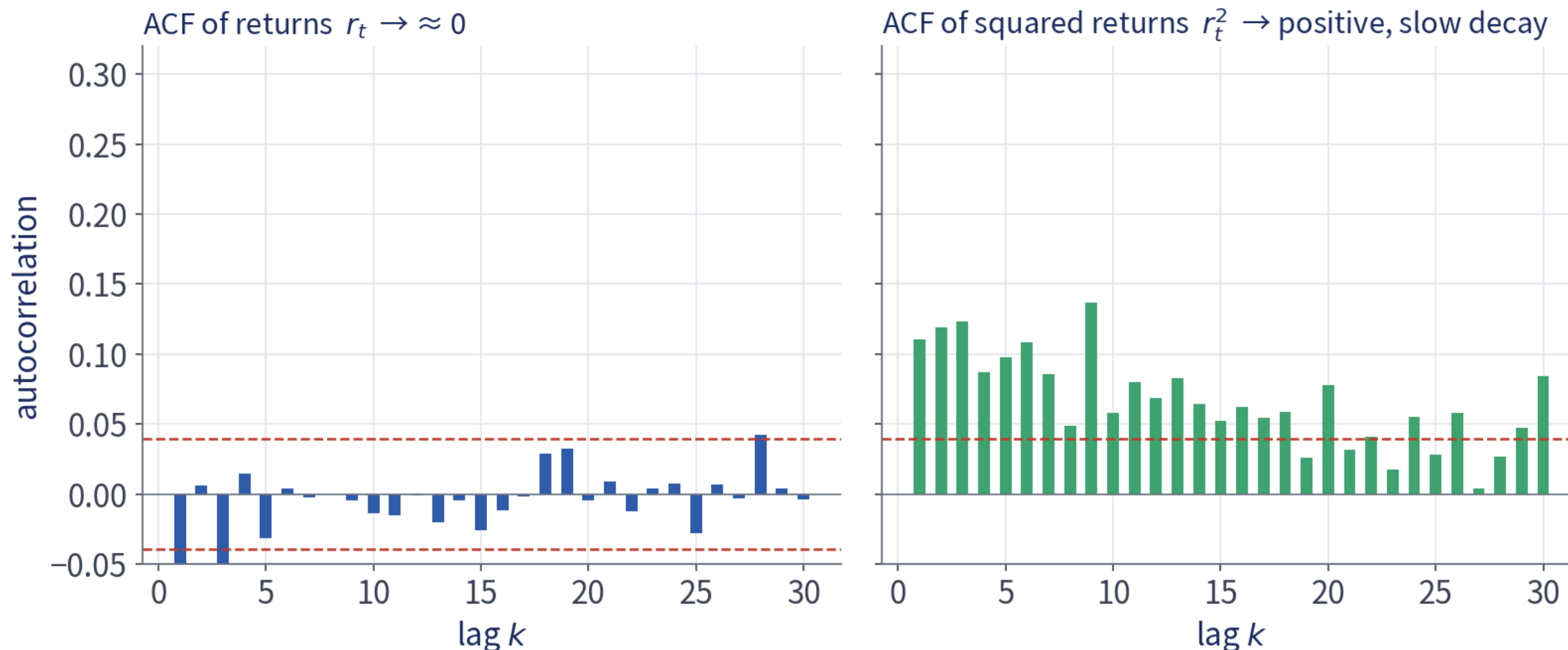
Volatility Clustering, Made Visible

Notice the alternating regimes: stretches of small wiggles, then bursts of large swings. This structure is absent from i.i.d. normal data.



The Split: Returns vs Squared Returns

Left: returns are uncorrelated. Right: squared returns are strongly, slowly-decaying autocorrelated — volatility has long memory.



Returns Are Unpredictable; Volatility Is Not

The single most important split in the course

$$\underbrace{\rho_R(k) \approx 0}_{\text{returns: no memory}} \quad \underbrace{\rho_{R^2}(k) > 0, \text{ slow decay}}_{\text{volatility: long memory}}$$

The level (returns)

- No linear memory — you cannot forecast the sign.
- This is the empirical face of market efficiency.
- Direction bets have no statistical edge from the past.

The magnitude (volatility)

- Strong, persistent memory — you CAN forecast risk.
- This is what GARCH-type models exploit (Week 2+).
- Risk management lives entirely on this side.

The Leverage Effect

Bad news bites harder than good news

$$\text{Corr}(r_t, \sigma_{t+1}^2) < 0 \quad (\text{leverage effect})$$

What it says

- A negative return raises future volatility more than a positive one of equal size.
- So returns and future volatility are negatively correlated.

Why the name

- A price drop raises a firm's debt-to-equity ratio ...
- ... making its equity riskier, hence more volatile.
- Behavioral fear effects reinforce the same pattern.

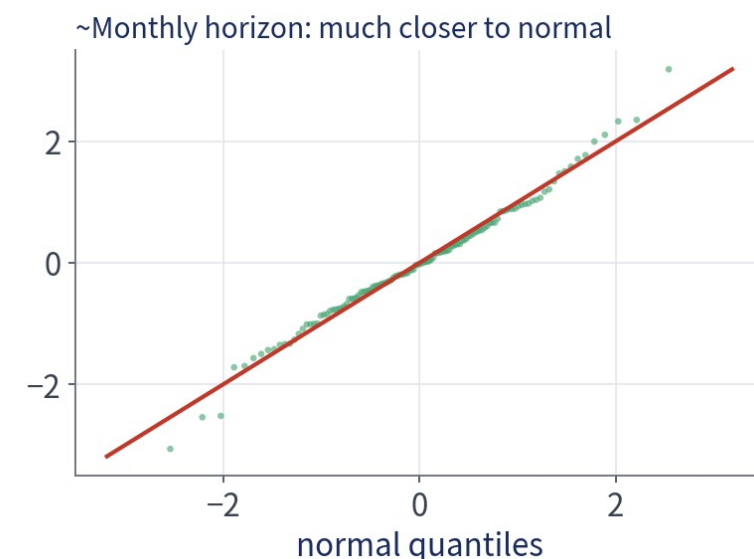
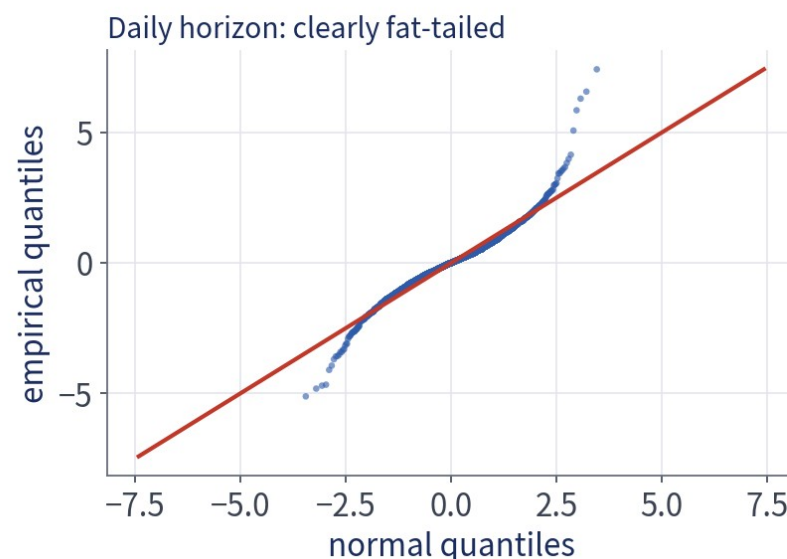
Aggregational Gaussianity

As the horizon lengthens, returns look more and more normal.

What to notice

- Daily: strongly fat-tailed.
- Monthly: nearly normal.
- Summing many returns ...
- ... pulls toward a normal.
- So the horizon matters.

Aggregational Gaussianity: longer horizons look more normal



The Six Stylized Facts Together

Three describe the shape of the distribution; three describe its dynamics through time. Every model is judged by how many it reproduces.

The Six Stylized Facts a Model Must Explain

Blue = dynamics through time | Green = shape of the distribution

1 No autocorrelation in returns

Returns are not linearly predictable.

2 Fat tails (excess kurtosis)

Extremes are far likelier than a normal predicts.

3 Gain / loss asymmetry

Crashes are sharper than rallies (negative skew).

4 Volatility clustering

Big moves cluster; today's vol predicts tomorrow's.

5 Leverage effect

Bad news raises future volatility more than good.

6 Aggregational Gaussianity

Longer-horizon returns look more normal.

From Facts to Models

Each fact motivates a specific tool later in the course

Fat tails

→ heavy-tailed

Replace the normal with a Student-t innovation.

Vol. clustering

→ GARCH

Let today's variance depend on yesterday's shock and variance.

Leverage

→ GJR / EGARCH

Let negative shocks raise variance more than positive ones.

No return memory

→ mean model

Keep the mean simple; put the structure in the variance.

The rest of the course is, in large part, a toolkit for reproducing these six facts.

Three Misconceptions to Avoid

Where intuition quietly goes wrong

Common myths

- "Zero autocorrelation means returns are independent."
- "Volatility clustering lets me predict which way prices go."
- "Fat tails are about a big average return."

The reality

- Uncorrelated $\neq$ independent — the squares are still dependent.
- You can forecast the SIZE of moves, never their SIGN.
- Tails are about the odds of extremes, not the center.

Hunting the Stylized Facts

You describe the goal; Claude designs and runs the exact tests

GOAL Confirm the six stylized facts in your own data using a META-PROMPT — you state what to check, Claude picks the methods.

[INPUT] Meta-prompt · Claude Code

```
Using returns.csv from Lab 1, write and run
stylized_facts.py that checks the facts we
covered this part:
- skewness & kurtosis vs the normal's 0 and 3;
- histogram and Q-Q plot vs a fitted normal;
- ACF of returns vs ACF of squared returns;
- how kurtosis changes at weekly / monthly
  horizons (aggregational Gaussianity).
For each, print the number AND one sentence on
what it implies, then write a short markdown
'stylized-facts report'.
```

Why a meta-prompt

- You give the GOAL; Claude chooses each test.
- Check every result against Part 2's six facts.
- Ask Claude to explain any test you don't know.
- Then refine: 'add a Student-t fit', etc.

Deliverable: stylized_facts.py + a one-page stylized-facts report on your asset.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1

Daily returns show ~zero autocorrelation but squared returns do not. Contradiction?

No — vol clusters

Q2

Your risk model assumes normal returns. What is the main danger?

Understates tails

Q3

Can volatility clustering help you predict tomorrow's return direction?

No — only size

All three follow directly from the six stylized facts.

Three Things to Carry Forward

Part 2 in one screen

1

Returns are not normal, and not i.i.d.

They have fat tails, negative skew, and volatility that clusters. The normal-i.i.d. picture fails on every count.

2

Remember the split: direction vs magnitude

Return direction is unpredictable (efficiency); return magnitude is predictable (volatility has memory). This split organizes the whole course.

3

Facts are the scoreboard for models

Fat tails → Student-t; clustering → GARCH; leverage → GJR/EGARCH. A model earns its keep by reproducing these facts.

Next — Part 3: Random Walks, Martingales, and the Efficient Market Hypothesis.